Unit S1: Statistics 1

Optional unit for IAS Mathematics and Further Mathematics
Optional unit for IAL Mathematics and Further Mathematics

Externally assessed

S1.1 Unit description

Mathematical models in probability and statistics; representation and summary of data; probability; correlation and regression; discrete random variables; discrete distributions; the Normal distribution.

S1.2 Assessment information

1. Examination

- First assessment: June 2019.
- The assessment is 1 hour and 30 minutes.
- The assessment is out of 75 marks.
- Students must answer all questions.
- Calculators may be used in the examination. Please see *Appendix 6: Use of calculators*.
- The booklet *Mathematical Formulae and Statistical Tables* will be provided for use in the assessments.

2. Notation and formulae

Students will be expected to understand the symbols outlined in *Appendix 7: Notation*.

Formulae that students are expected to know are given below and will **not** appear in the booklet *Mathematical Formulae and Statistical Tables*, which will be provided for use with the paper. Questions will be set in SI units and other units in common usage.

Mean $= \bar{x} = \dfrac{\sum x}{n}$ or $\dfrac{\sum fx}{\sum f}$

Standard deviation $= \sqrt{(\text{variance})}$

Interquartile range = IQR = $Q_3 - Q_1$

$P(A') = 1 - P(A)$

For independent events A and B,

$P(B \mid A) = P(B)$, $P(A \mid B) = P(A)$,

$P(A \cap B) = P(A)\,P(B)$

$E(aX + b) = aE(X) + b$

$\text{Var}(aX + b) = a^2\,\text{Var}(X)$

2. Notation and formulae *continued*

Cumulative distribution function for a discrete random variable:

$$F(x_0) = P(X \leqslant x_0) = \sum_{x \leqslant x_0} p(x)$$

Standardised Normal Random Variable $Z = \dfrac{X - \mu}{\sigma}$

where $X \sim N(\mu, \sigma^2)$

S1.3 Unit content

What students need to learn:	Guidance